

$$\begin{aligned}
\mathbf{1. \quad a.} \quad \det \begin{bmatrix} -1 & 3 & 5 \\ 6 & 4 & 2 \\ -2 & 5 & 1 \end{bmatrix} &= \det \begin{bmatrix} -1 & 3 & 5 \\ 0 & 22 & 32 \\ 0 & -1 & -9 \end{bmatrix} = -\det \begin{bmatrix} -1 & 3 & 5 \\ 0 & -1 & -9 \\ 0 & 22 & 32 \end{bmatrix} = -\det \begin{bmatrix} -1 & 3 & 5 \\ 0 & -1 & -9 \\ 0 & 22 & 32 \end{bmatrix} = \\
&-\det \begin{bmatrix} -1 & 3 & 5 \\ 0 & -1 & -9 \\ 0 & 0 & -166 \end{bmatrix} = -(-1)(-1)(-166) = 166 \\
\mathbf{b.} \quad \det \begin{bmatrix} 1 & -1 & 0 & 1 \\ 0 & 2 & 1 & 1 \\ 2 & -2 & 2 & 3 \\ 0 & 0 & 6 & 2 \end{bmatrix} &= \det \begin{bmatrix} 1 & -1 & 0 & 1 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 6 & 2 \end{bmatrix} = \det \begin{bmatrix} 1 & -1 & 0 & 1 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix} = (1)(2)(2)(-1) = \\
&-4
\end{aligned}$$

- 2.** A is square. So if A has a row of zeros then it is singular which implies that $\det A = 0$ by Theorem 1.2.

Alternatively, Suppose that the j^{th} row is all zeros, $\mathbf{A}_j = \mathbf{0}$. Then $2\mathbf{A}_j = \mathbf{A}_j = \mathbf{0}$. Now apply property (2) of proposition 1.1 to the j^{th} row so that $\det A = 2 \det A \Rightarrow \det A = 0$.

Note that the same result can be shown for a matrix with a column of zeros using Corollary 1.8.

- 5.** cA means that c multiplies each of the n rows of A . So then by property (2) of proposition 1.1, $\det(cA) = \underbrace{c \cdot c \cdots c}_n \det A = c^n \det A$.

- 9.** We need to assume the results from parts (b). If $k = \ell$, then from part (b),

$$\det \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] = \det A \det(D - CA^{-1}B).$$

By Theorem 1.5, $\det A \det(D - CA^{-1}B) = \det(AD - ACA^{-1}B)$. It is given that $CA = AC$ (this assumption is necessary) so then

$$\begin{aligned}
\det A \det(D - CA^{-1}B) &= \det(AD - (AC)A^{-1}B) \\
&= \det(AD - (CA)A^{-1}B) \\
&= \det(AD - CB)
\end{aligned}$$

as needed.